

Learning Controlled Stochastic Differential Equations

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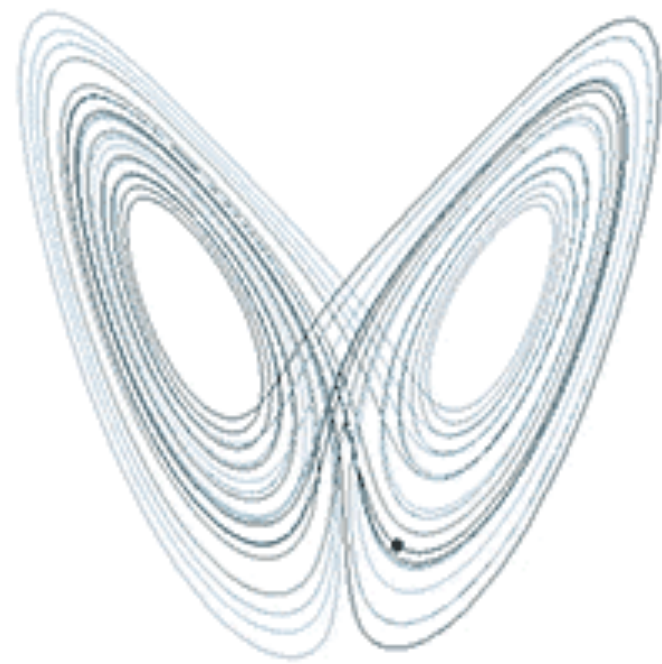
³ SDA Bocconi

WCCM-ECCOMAS, July 2026

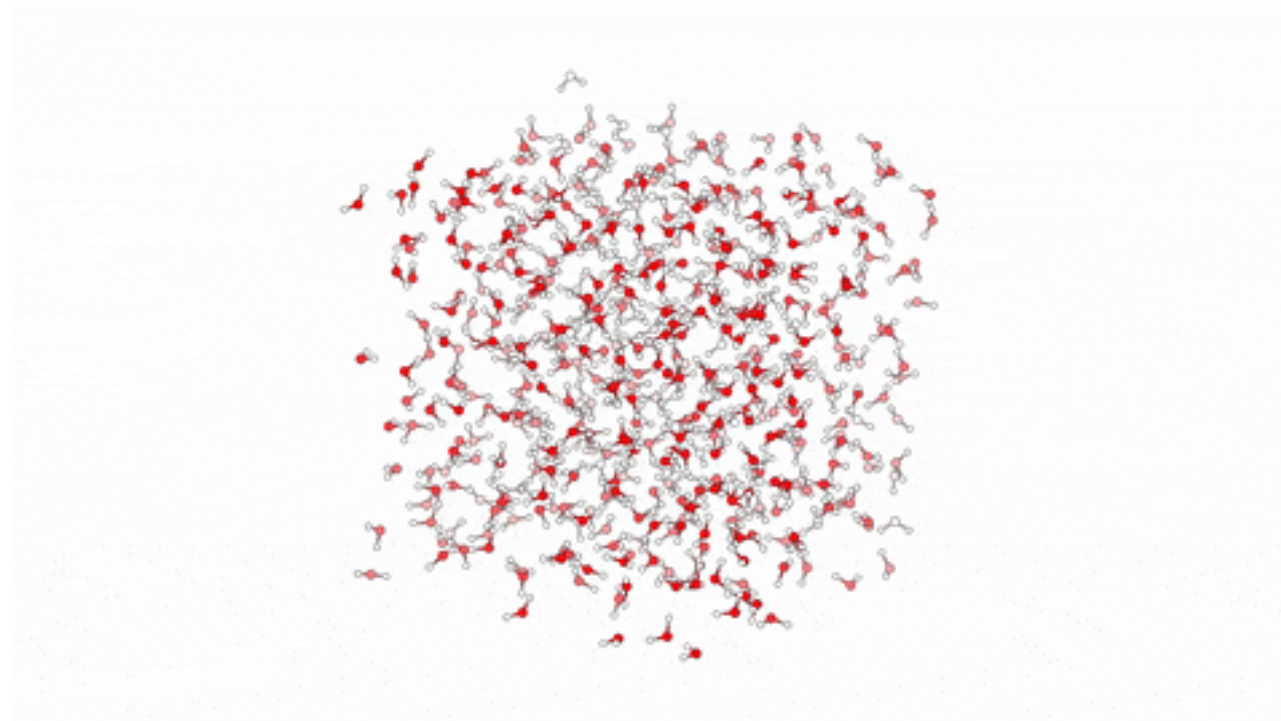
Controlled Stochastic Dynamical Systems

Dynamical systems:

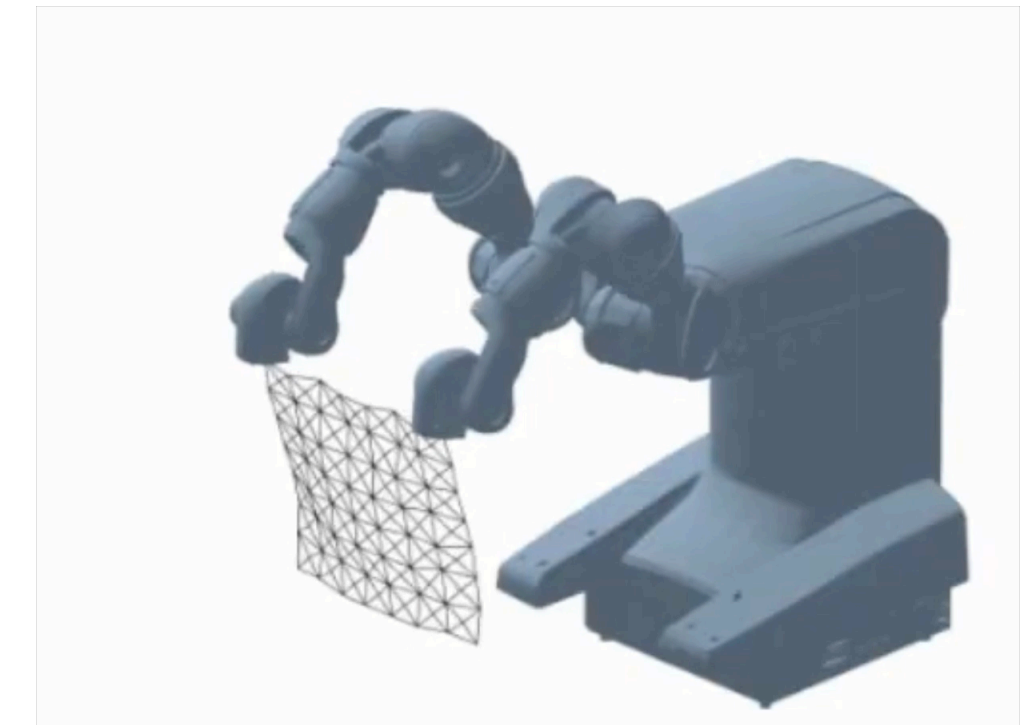
Lorenz 63



Molecular dynamics



Robotic arm [1, 2]



Models of dynamical systems:

Continuous-time

$x_t \in \mathbb{R}^n$: system's state at time t

$$dx_t = b(x_t) dt$$

Stochastic

W_t : Brownian motion

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t$$

drift

diffusion

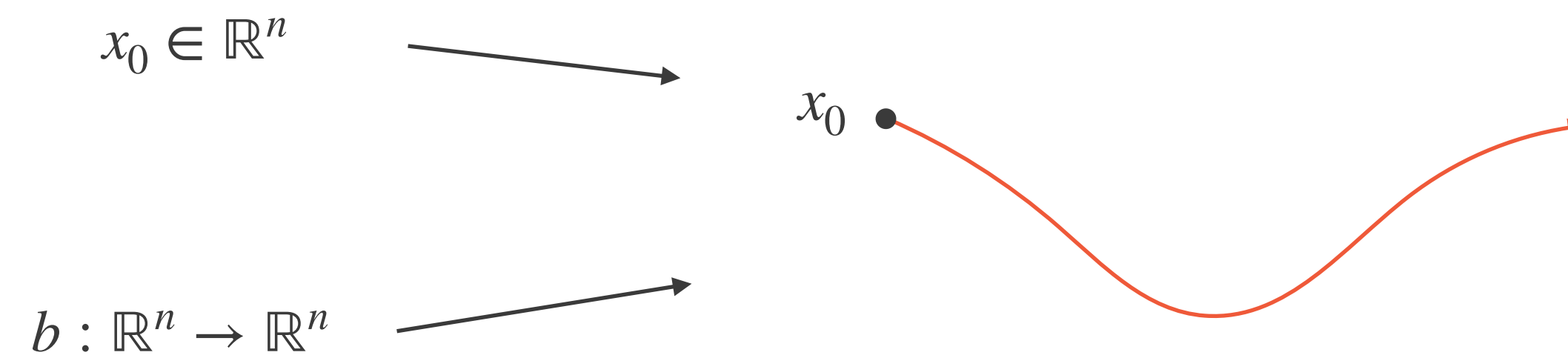
Controlled

$u_t \in \mathbb{R}^d$: control input at time t

$$dX_t = b(X_t, u_t) dt + \sigma(X_t, u_t) dW_t$$

Why models of controlled dynamical systems?

Predict system's evolution:



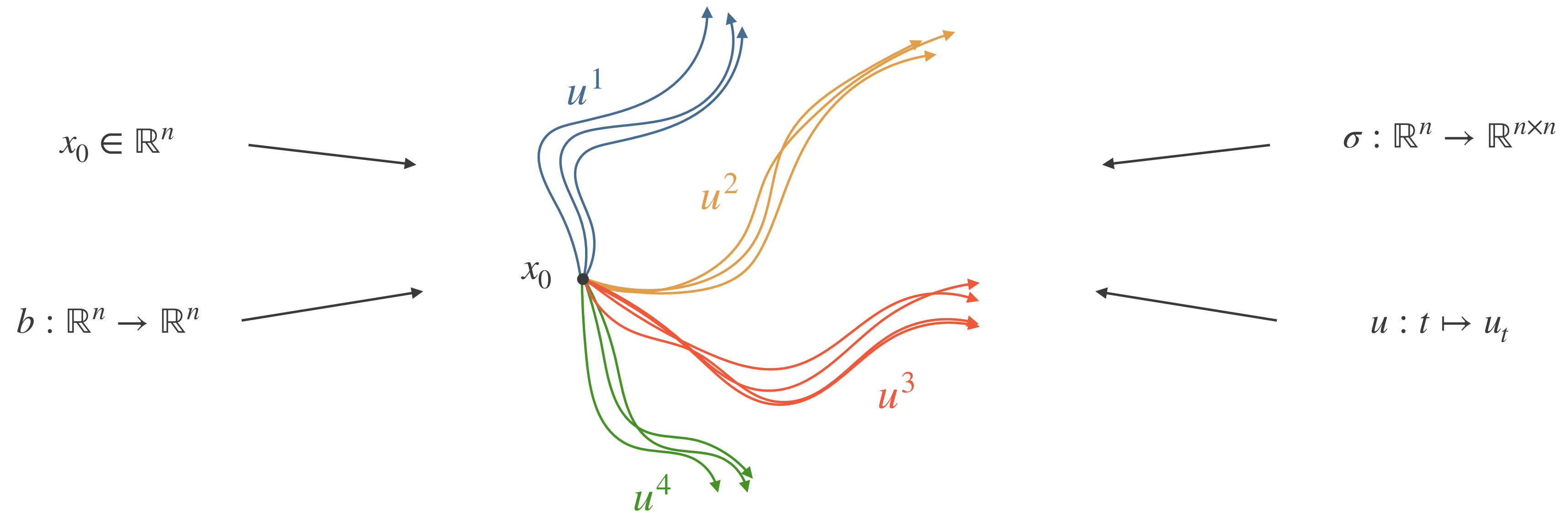
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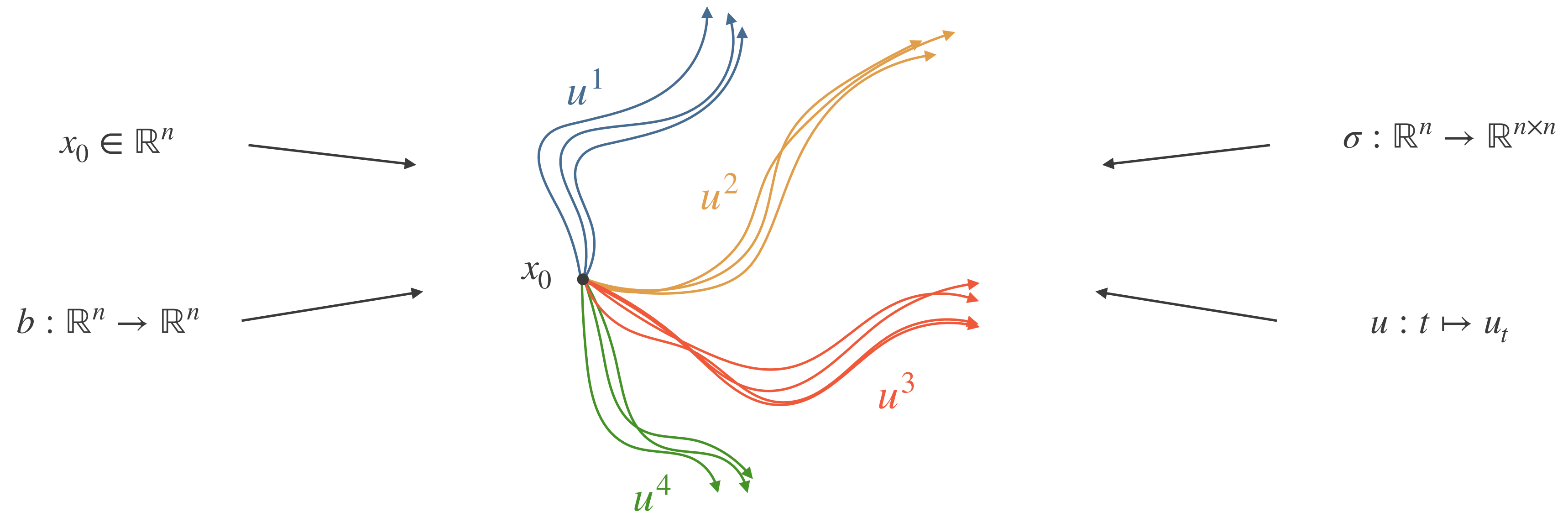
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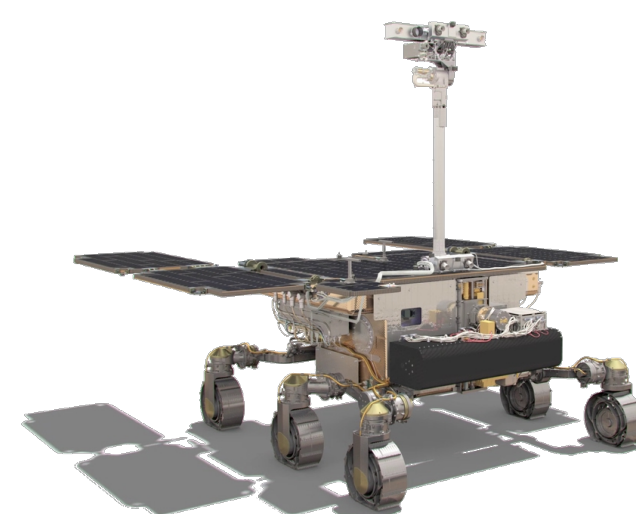


Numerous applications: design controls to achieve desired behavior

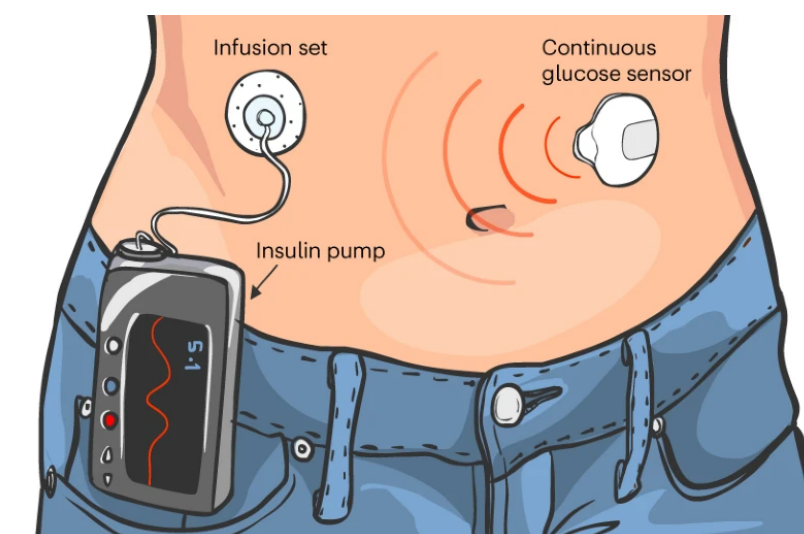
Industrial processes



Robotic systems



Medical systems [3]



Learning problem

In many applications, the model is unknown:

$$(b, \sigma) = ?$$

Learning problem

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$$(b, \sigma) = (b_0, \sigma_0) + ?$$

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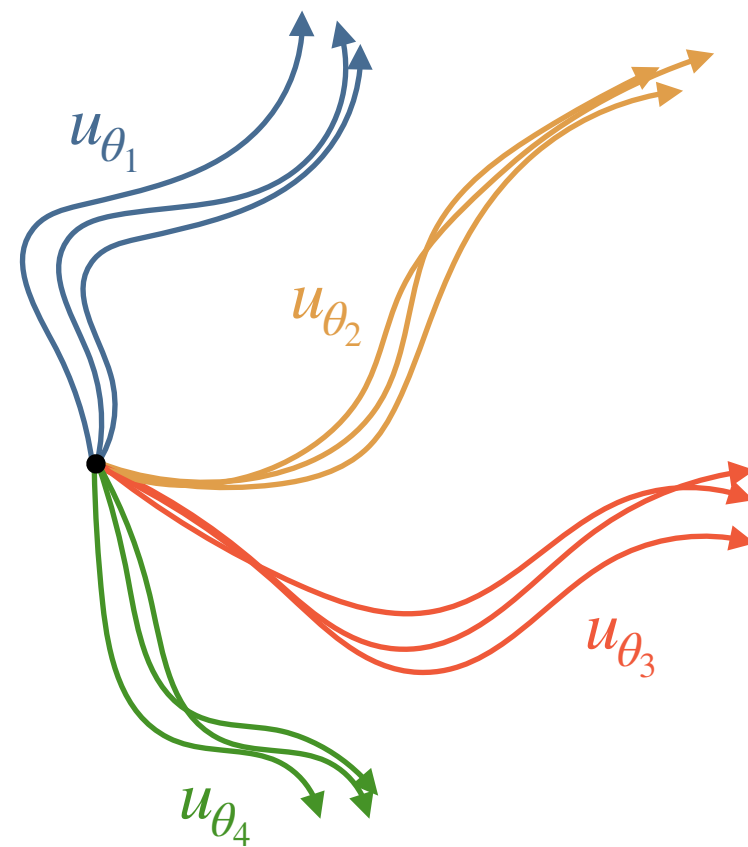
Parametrized control space:

$$u_\theta : [0, T] \rightarrow \mathbb{R}^d$$

$$\theta \in \Theta \subset \mathbb{R}^m$$

bounded set

Controlled SDE estimation: learning (b, σ) by collecting trajectories under various controls $u_{\theta_1}, u_{\theta_2}, \dots, u_{\theta_K}$



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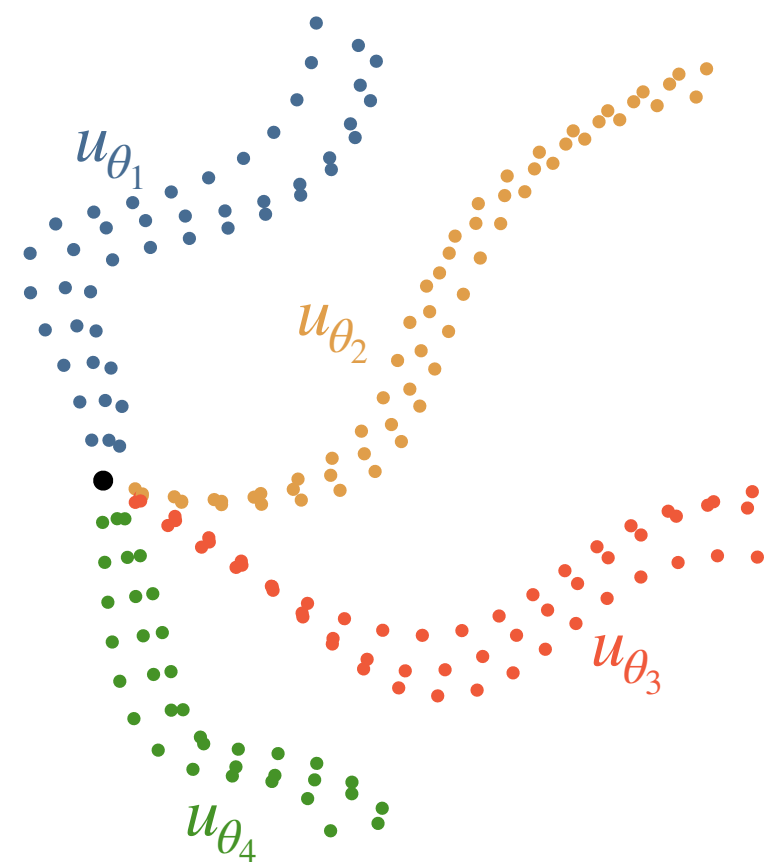
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$$\{(\theta_k, X_{\theta_k}(w_i^k, t_l))\}_{k,i,l} \in \mathbb{R}^{K \times N \times M}$$

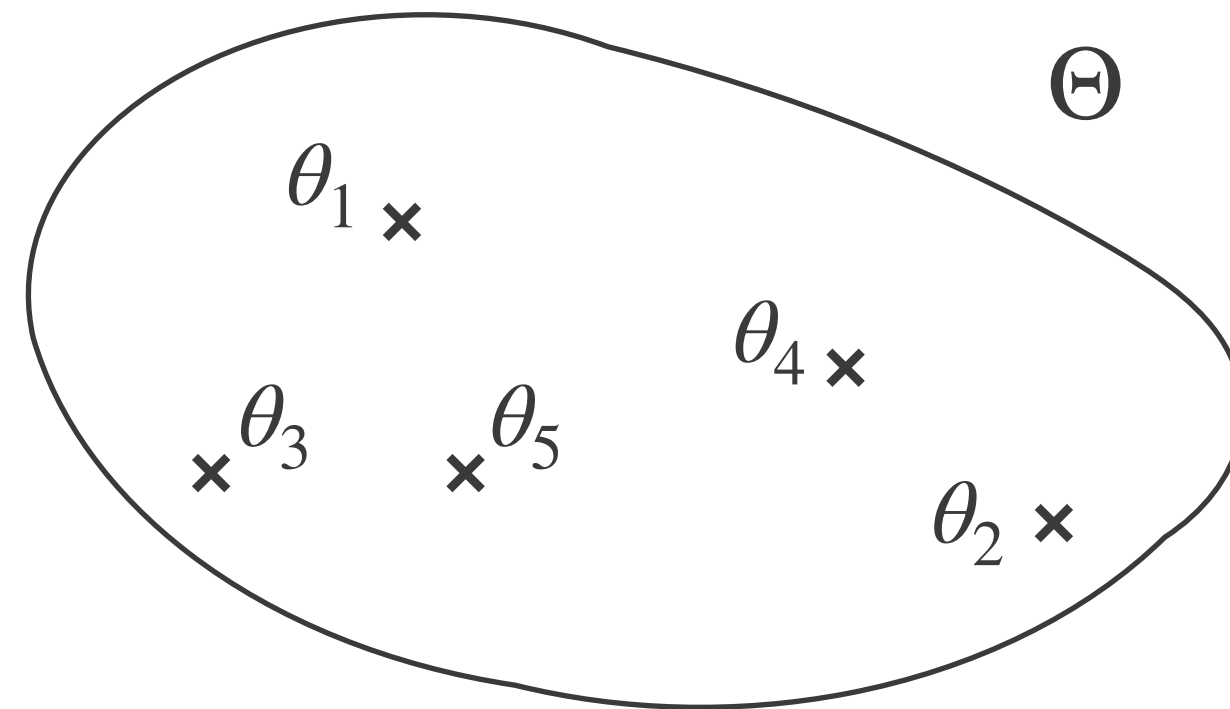
$$K = 4, N = 3, M \approx 25$$

Focus of this work

Density flows: probability density of $X_\theta(t)$ under control u_θ

$$\theta, t, x, \mapsto p_{b,\sigma}(\theta, t, x)$$

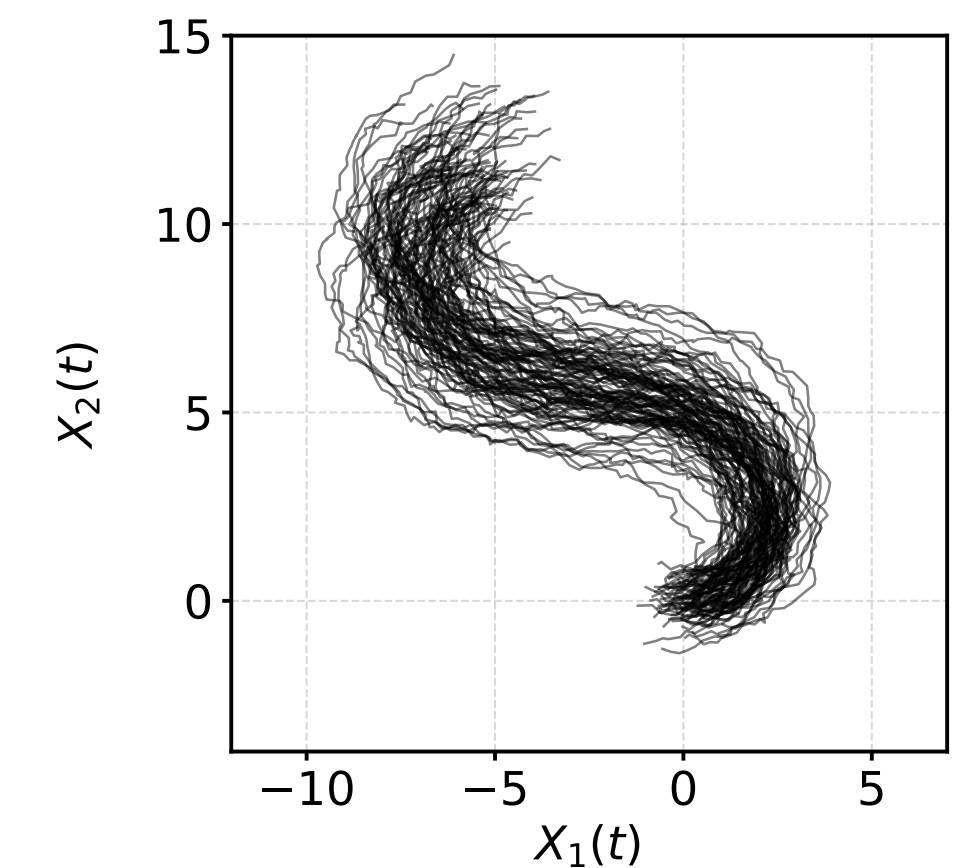
Focus: reproduce the density flow across controls



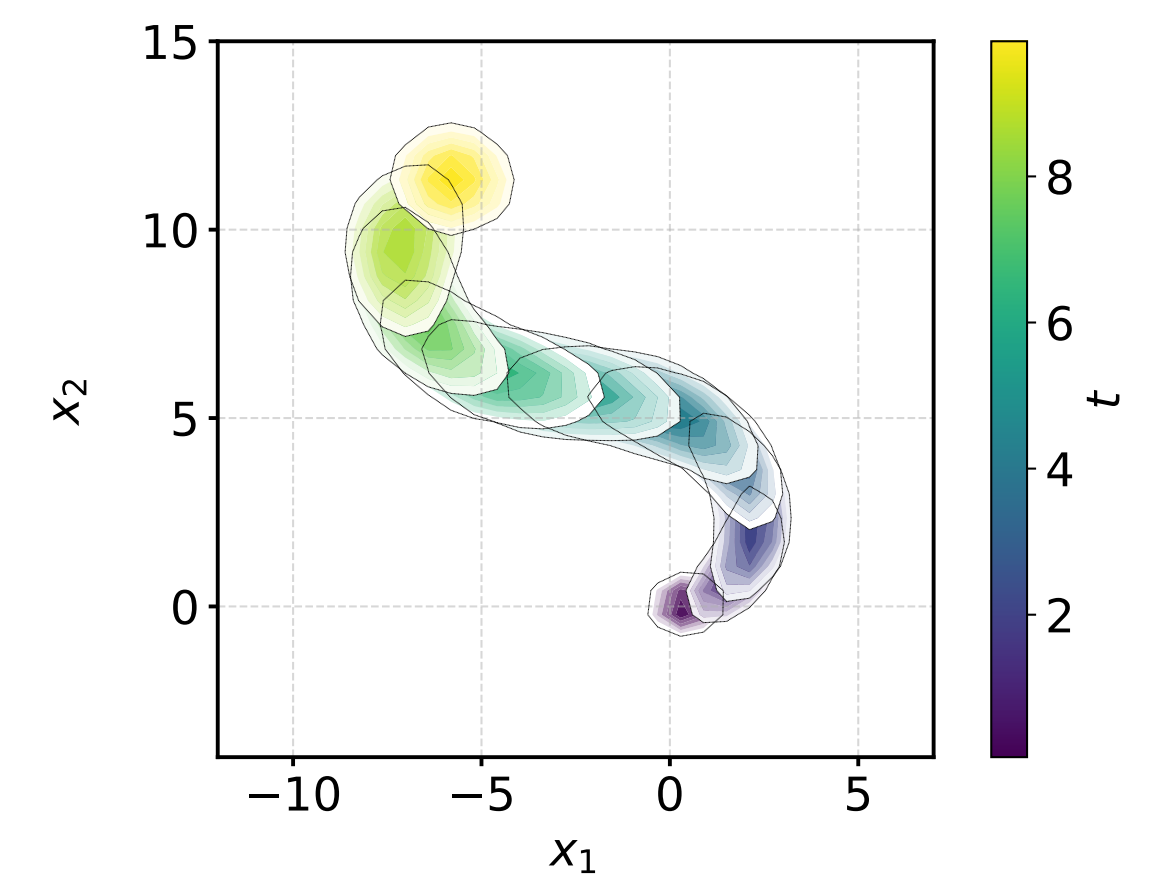
$$\forall \theta \in \Theta, \quad p_{\hat{b},\hat{\sigma}}(\theta) \approx p_{b,\sigma}(\theta)$$

Remark: different drift–diffusion pairs may induce the same density flow

Trajectories under u_θ



Density flow under u_θ



The Fokker-Planck equation

Fokker-Planck equation: let $X_\theta(t)$ governed by

$$dX_t = b(t, X_t, u_\theta(t)) dt + \sigma(t, X_t, u_\theta(t)) dW_t$$

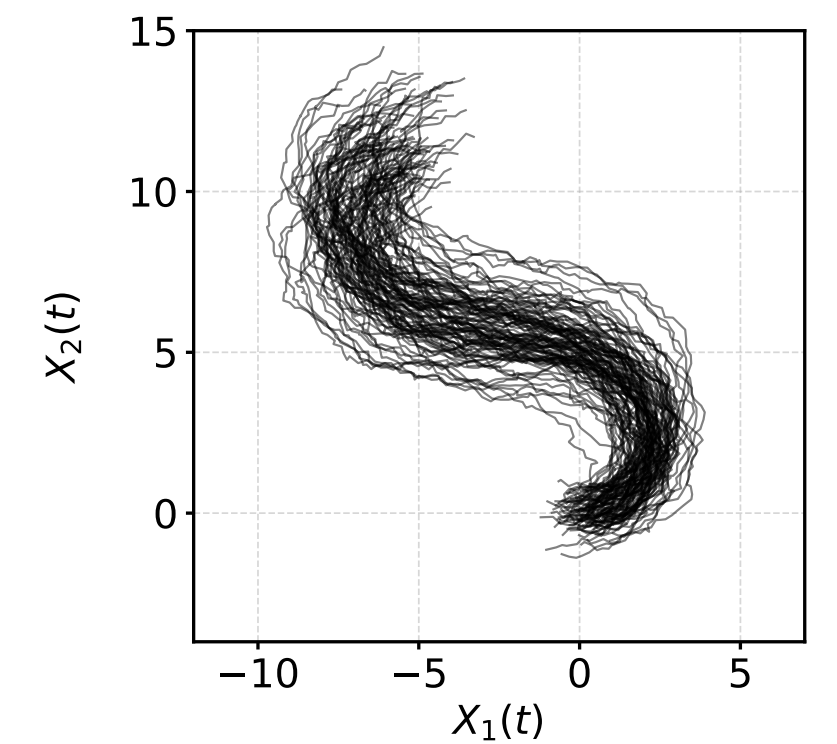
then under suitable regularity assumptions, its density flow p satisfies

$$\partial_t p(\theta, t, x) = L^{(b,a),\theta} p(\theta, t, x),$$

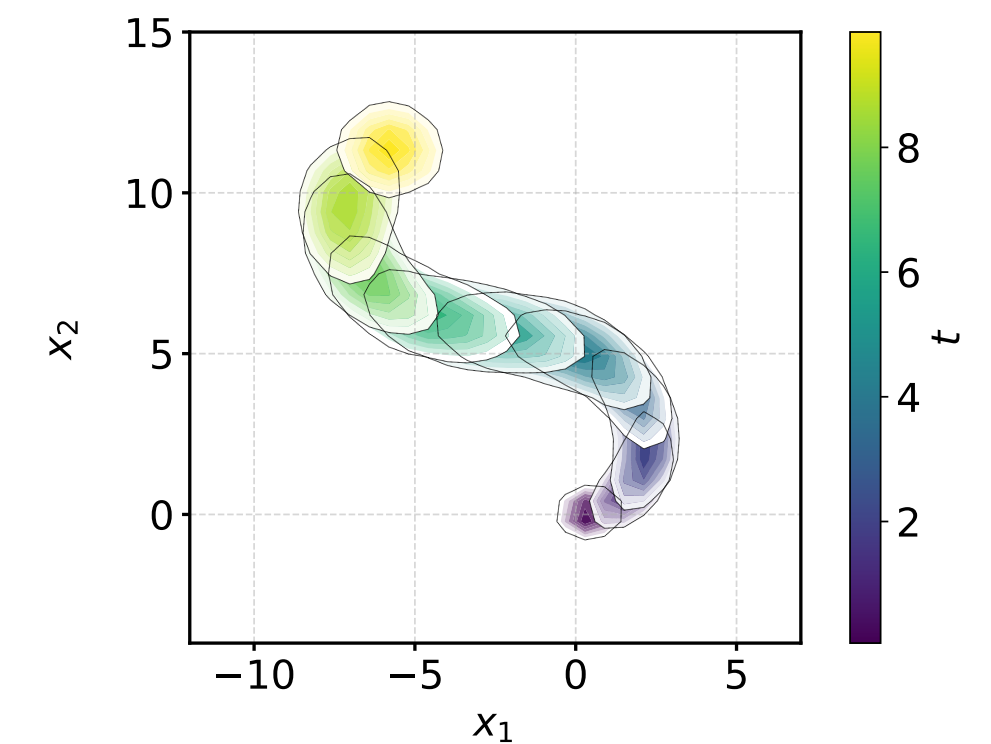
where

$$a = \sigma\sigma^T, \quad L^{(b,a),\theta} q(t, x) = \frac{1}{2} \sum_{i,j=1}^n \partial_{x_i x_j} \left(a_{ij}(t, x, u_\theta(t)) q(t, x) \right) - \sum_{i=1}^n \partial_{x_i} \left(b_i(t, x, u_\theta(t)) q(t, x) \right)$$

Trajectories under u_θ



Density flow under u_θ



Proposed estimator

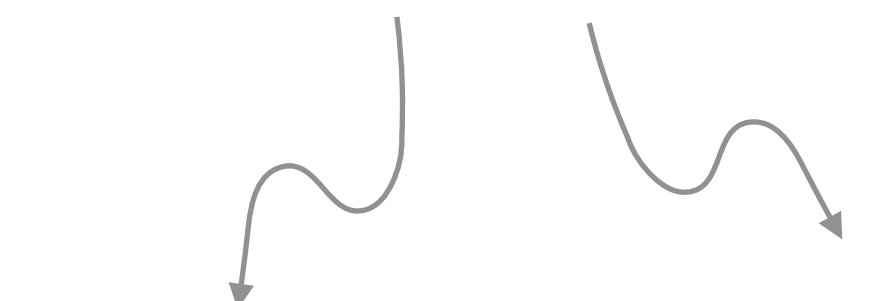
Two-step method:

1. Estimate the density flows $\hat{p}_k \approx p(\theta_k, \cdot, \cdot)$ for each of the controls in the dataset $u_{\theta_1}, \dots, u_{\theta_K}$
2. Estimate coefficients thanks to the Fokker–Planck equation:

$$\arg \min_{(\hat{b}, \hat{\sigma}) \in \mathcal{F}} \left\| \frac{\partial p_{b, \sigma}}{\partial t} - L^{\hat{b}, \hat{\sigma}} p_{b, \sigma} \right\|_{L^2}^2$$

Model $\mathcal{F}$ for the coefficients:

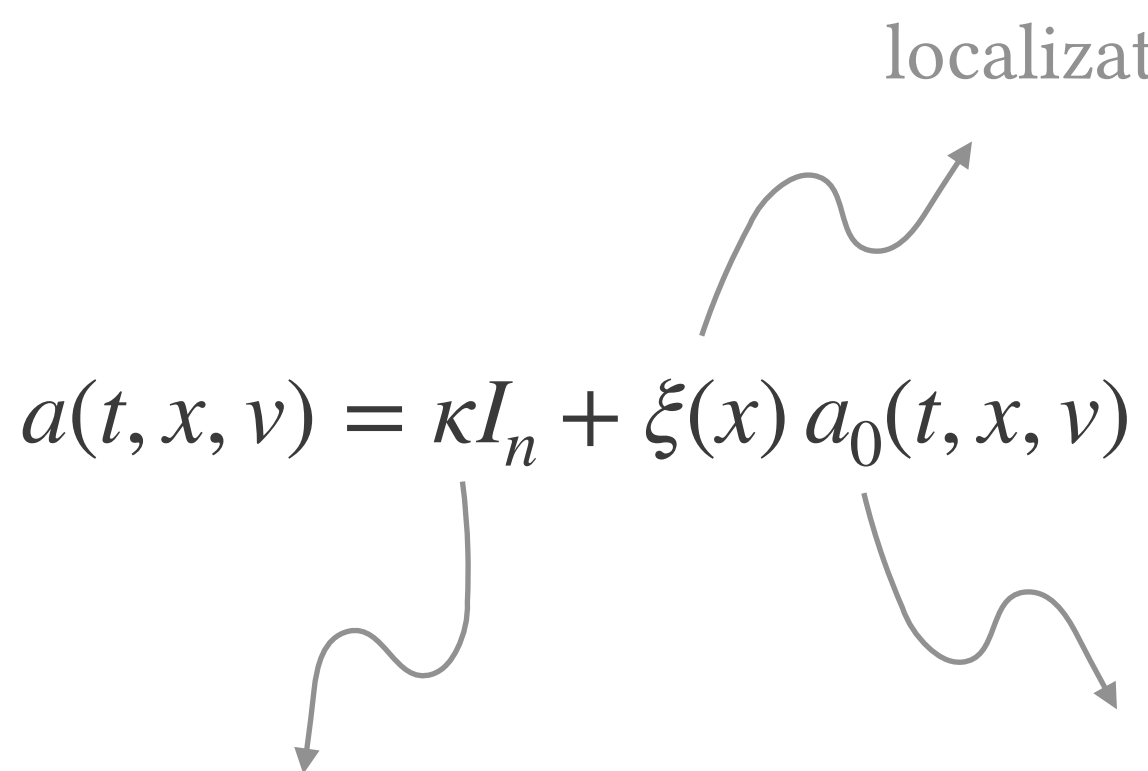
$$b(t, x, v) = \xi(x) b_0(t, x, v)$$



 localization RKHS model

$$a = \sigma \sigma^T$$

$$a(t, x, v) = \kappa I_n + \xi(x) a_0(t, x, v)$$



 localization
 ellipticity floor positive-semidefinite RKHS model [4]

Main theoretical guarantees

Statistical guarantee: under sufficiently accurate density flow estimations, with probability at least $1 - \delta$

$$\left\| p_{\hat{b}, \hat{a}} - p_{b, a} \right\|_{L^2(\Theta \times [0, T] \times \mathbb{R}^n)} \lesssim \left(\frac{\log(K/\delta)}{\sqrt{K}} \right)^{\frac{1}{1 - \min(r, s)}}$$

with r, s measuring the regularity-dimension of the true controlled system in the state–time and control directions.

Example: for a ν -Sobolev-regular controlled system[★]

$$r < 1 - \frac{\dim(X) + 1}{2\nu}, \quad s < 1 - \frac{\dim(\Theta)}{2\nu}$$

Take-home message:

Higher regularity and lower dimension $\implies$ faster generalization

(★) $(\theta, t) \mapsto u_\theta(t)$, $(\theta, t, x) \mapsto p(\theta, t, x)$, $(t, x, v) \mapsto (b, a)(t, x, v)$ are sufficiently Sobolev-regular so that $(\theta, t, x) \mapsto L_{b, a}^{u_\theta} p(\theta, t, x)$ has H^ν -regular sections in θ and (t, x)

Beyond average performance?

Beyond average accuracy?

$L^2(\Theta)$ bounds give accuracy on average over controls

Uniform guarantees: under same assumptions, with probability at least $1 - \delta$

$$\left\| p_{\hat{b}, \hat{a}} - p_{b,a} \right\|_{L^\infty(\Theta; L^2([0, T] \times \mathbb{R}^n))} \lesssim \left(\frac{\log(K/\delta)}{\sqrt{K}} \right)^{\frac{r \wedge s}{1 - r \wedge s}}$$

slower than L^2

Tail-risk guarantees: if $\mathbb{E}[\|X_0\|^\beta] < \infty, \beta > 2$, with probability at least $1 - \delta$

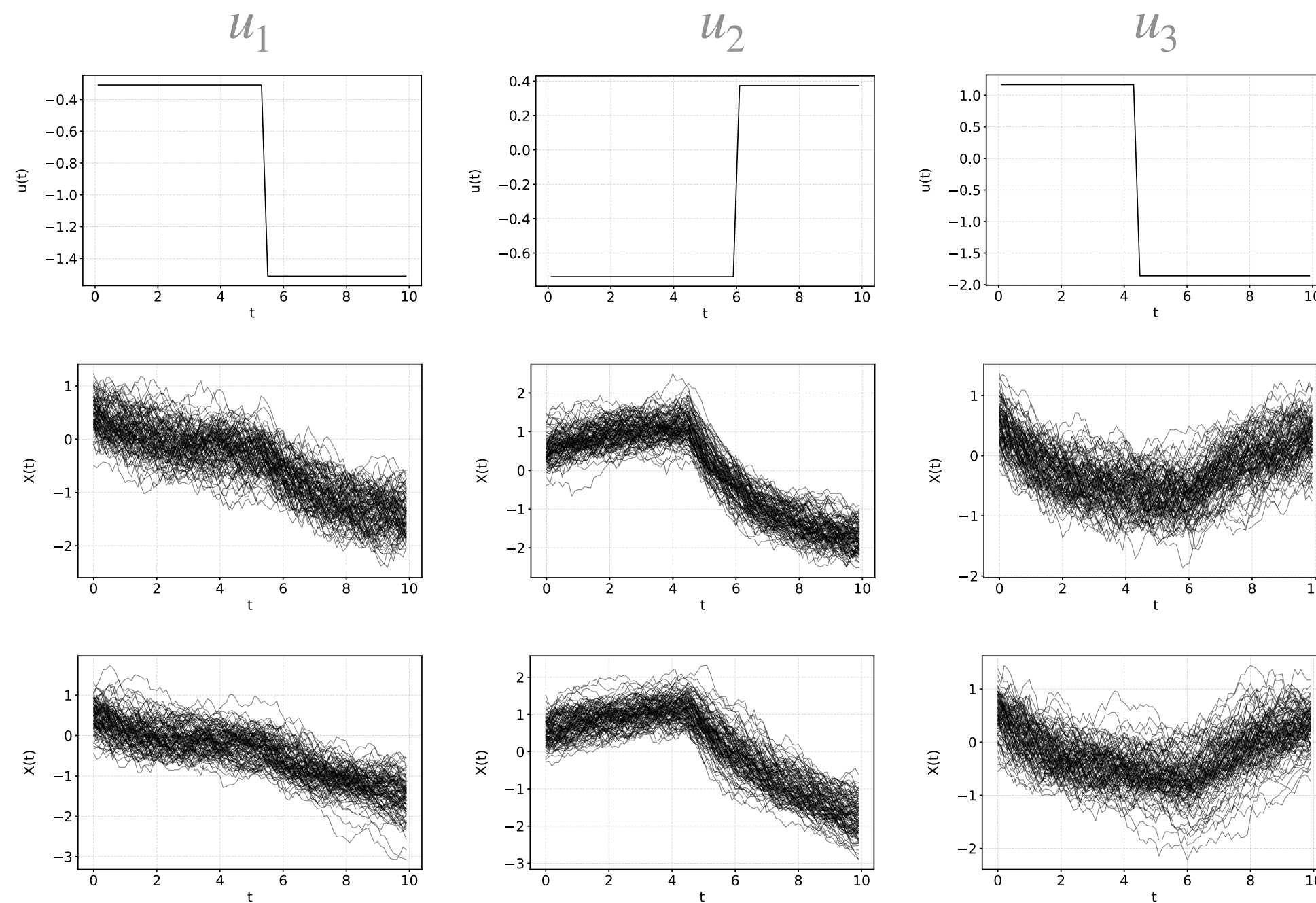
$$\sup_{\theta \in \Theta} \left\| \text{CVaR}_\rho(f(X_{\hat{b}, \hat{a}}^\theta(\cdot))) - \text{CVaR}_\rho(f(X_{b,a}^\theta(\cdot))) \right\|_{L^2(0, T)} \lesssim \frac{\|f\|_\infty}{\rho} \left(\frac{\log(K/\delta)}{\sqrt{K}} \right)^{\frac{r \wedge s}{1 - r \wedge s} \frac{\beta}{\beta + n/2}}$$

slower than L^∞

Numerical illustration

Ornstein–Uhlenbeck process: training on $K = 10$ controls, $N = 1000$ trajectories.

$$dX(t) = \frac{1}{2}(u(t) - X(t)) dt + \frac{1}{\sqrt{8}} dW(t), \quad u(t) = \begin{cases} \theta_0 & t < \theta_3 \\ \theta_1 & t \geq \theta_3 \end{cases}$$



Test controls

True trajectories from (b, a)

Simulated trajectories from $(\hat{b}, \hat{a})$

Takeaways and perspectives

Takeaways:

- Learn controlled dynamics from finitely many controls
- Reproduce density flow through the *Fokker-Planck equation*
- Obtain statistical convergence rates governed by regularity and dimension

Code:



lmotte/sde-learn

lmotte/controlled-sde-learn

Perspectives:

- From open-loop to feedback controls
- From exact kernels to scalable approximations
- From simulations to experiments on real systems



References

- (1) Amadio, Fabio, et al. "Controlled Gaussian process dynamical models with application to robotic cloth manipulation: F. Amadio et al." *International Journal of Dynamics and Control* 11.6 (2023): 3209-3219.
- (2) Caldarelli, Edoardo, et al. "Dynamic robotic cloth folding with efficient Koopman operator-based model predictive control." *arXiv preprint arXiv:2605.18373* (2026).
- (3) Khan, M. (2024). "Navigating Insulin Pumps: Technology, Benefits, and What's Next." *News-Medical.Net*.
- (4) Muzellec, Boris, Francis Bach, and Alessandro Rudi. "Learning PSD-valued functions using kernel sums-of-squares." *arXiv preprint arXiv:2111.11306* (2021).